

CS & IT ENGINEERING



COMPUTER ORGANIZATION AND ARCHITECTURE

CPU & Control Unit

Lecture No.- 1

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Recap of Previous Lecture



Topic

Questions on Addressing Modes

Topics to be Covered



Topic

CPU

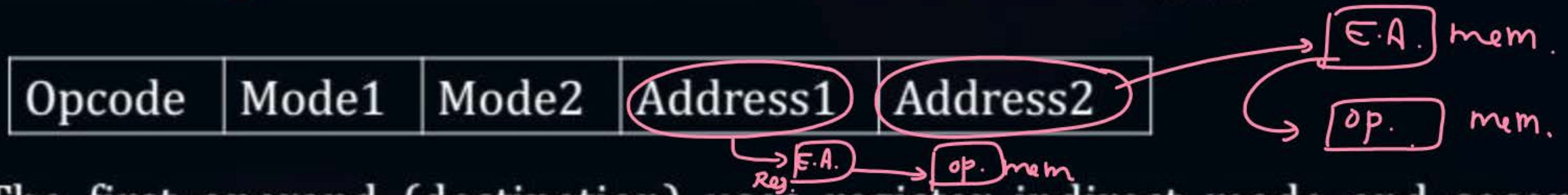
Topic

MIPS

Topic

Data Path

#Q. Consider a 6-words instruction, which is of the following type:



The first operand (destination) uses register indirect mode and second operand uses indirect mode. Assume each operand is of size 2 words, each address is of 2 words and main memory takes 50ns for 1 byte access. Further assume that the opcode denotes addition operation which copies result of addition of 2 operands. One word size is 4 bytes. Total time required in:

1. Fetch cycle of instruction $24 * 50 = 1200 \text{ ns}$
2. Execution cycle of instruction $= 1600 \text{ ns}$
3. Instruction cycle of instruction $= 2800 \text{ ns}$

2.

operand 1

operand 2

Decode

C.A. calculatⁿ &
operand fetch

8 * 50 ns

8 * 50 = 400
+ 8 * 50 = 400 } 800 ns

Execution

write back

8 * 50 ns

Total 1600 ns

New chapter

CPU



Topic : CPU



1. CPU Cycle :- Time in which CPU Can perform smallest micro-operation.

2. CPU Clock rate = $\frac{1}{\text{CPU cycle time}}$

3. CPI $\xrightarrow{\text{No. of CPU cycles needed to execute an instruction}}$
(cycles per instⁿ)

4. Execution Time
avg. instⁿ execution time = $\text{CPI}_{\text{avg}} * \text{cycle time}$

prog. execution time = $n * \text{CPI}_{\text{avg}} * \text{cycle time}$

$$\frac{1}{10^9} = \text{sec}$$

$$\frac{1}{10^6} = \text{micro}$$

$$\frac{1}{10^3} = \text{milli}$$

$$\text{prog. execution time} = \frac{n * CPI_{avg}}{\text{clock rate}}$$

$$avg. CPI = \frac{\text{Total no. of cycles to execute } n \text{ inst}^ns}{\text{Total no. of inst}^n}$$



Topic : Average CPI

Consider computing the overall CPI for a machine A for which the following performance measures were recorded when executing a set of benchmark programs. Assume that the clock rate of the CPU is 200 MHz

Instruction Category	Number of Instructions	No. of cycles per Instruction	Total cycles
ALU	48	2	$48 * 2 = 96$
Load & Store	10	5	$10 * 5 = 50$
Branch	39	4	$39 * 4 = 156$
Other	3	3	$3 * 3 = 9$
	100	Total	= 311

$$CPI = \frac{311}{100} = 3.11$$



Topic : MIPS

(million Inst^{ns} Per Second)



in t time CPU can execute no. of inst^{ns} = n

$$\text{in 1 time} \quad \frac{n}{t} = \frac{n}{t}$$

$$= \frac{\cancel{n} * \text{clock rate}}{\cancel{n} * \text{CPI}_{\text{avg}}}$$

$$= \frac{\text{clock rate}}{\text{CPI}_{\text{avg}}}$$

$$\text{MIPS} = \frac{\text{clock rate}}{\text{CPI}_{\text{avg}} * 10^6}$$

In prev. example

$$\text{clock rate} = 200 \text{ MHz}$$

$$\text{CPI}_{\text{avg}} = 3.11$$

$$\text{MIPS} = \frac{200 \cancel{\text{MHz}}}{3.11 * \cancel{10^6}}$$

$$= 64.3 \text{ MIPS}$$

#Q. Consider two processors P1 and P2 executing the same instruction set. Assume that under identical conditions, for the same input, a program running on P2 takes 25% less time but incurs 20% more CPI (clock cycles per instruction) as compared to the program running on P1. If the clock frequency of P1 is 1GHz, then the clock frequency of P2 (in GHz) is 1.6?

	P1	P2
CPI	C_1	$C_2 = 1.2 C_1$
clock freq.	$f_1 = 1\text{GHz}$	$f_2 = ?$
time	t_1	$t_2 = 0.75 t_1$
no. of inst ^{ns}	n_1	n_2

$$n_1 = n_2$$

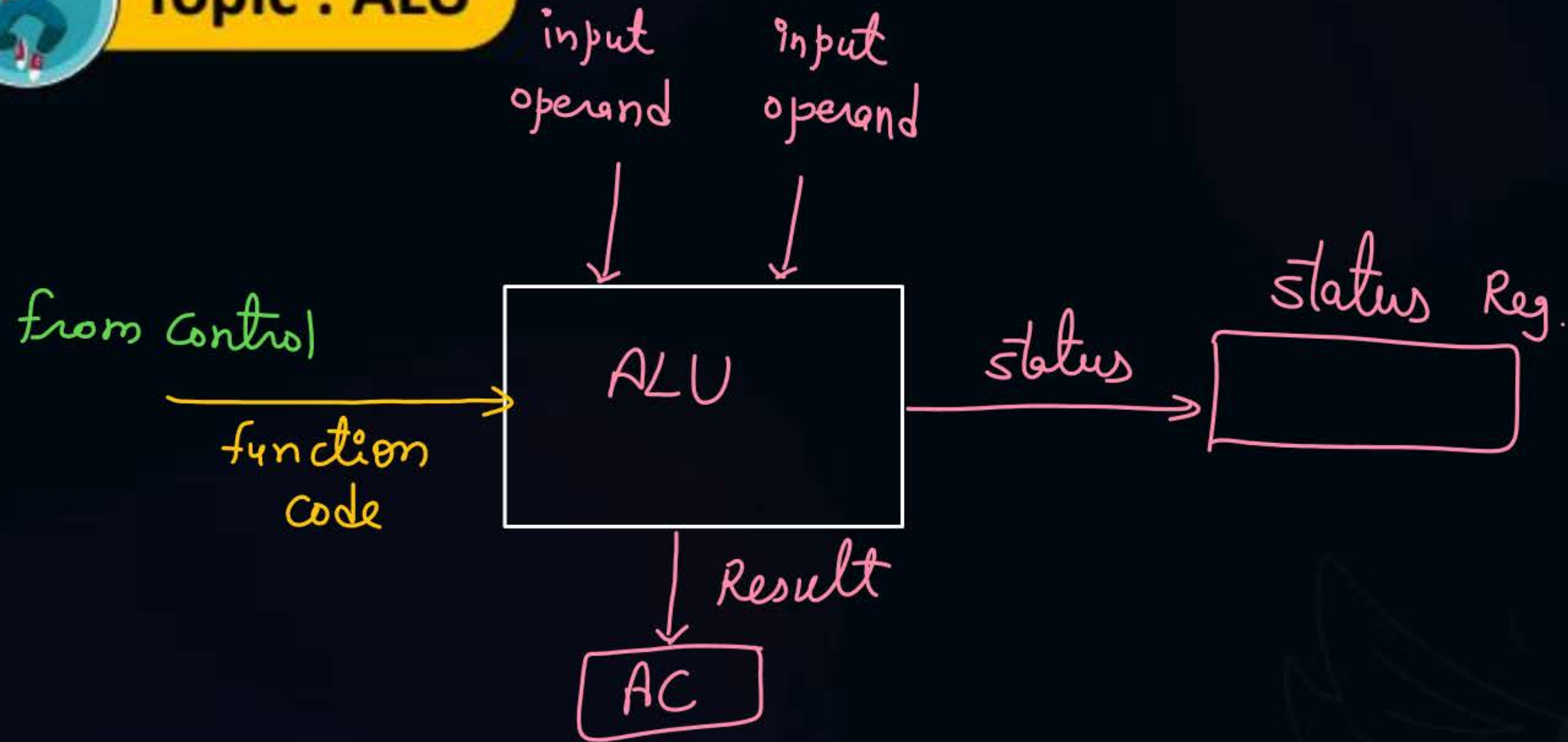
$$\frac{t_1 * f_1}{C_1} = \frac{t_2 * f_2}{C_2}$$

$$\frac{\cancel{t_1} * 1\text{GHz}}{\cancel{C_1}} = \frac{0.75 \cancel{t_1} * f_2}{1.2 \cancel{C_1}}$$

$$f_2 = \frac{1.2}{0.75} * 1\text{GHz} = 1.6\text{GHz}$$



Topic : ALU





Topic : Datapath

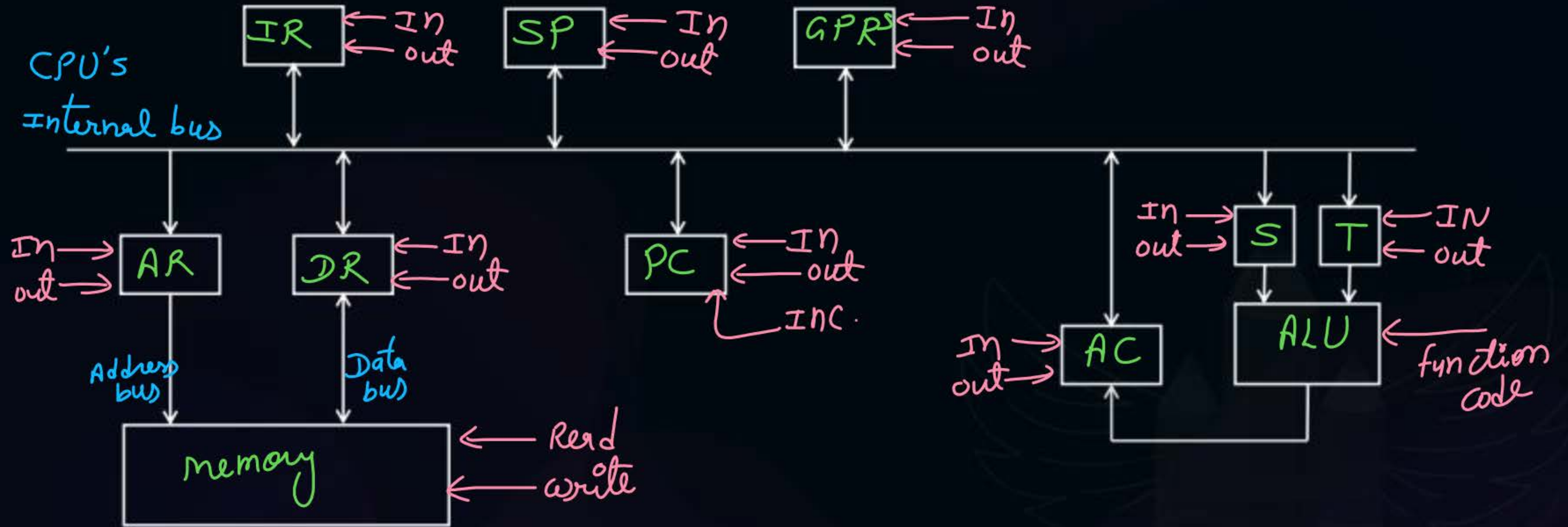
→ CPU design

Collection of functional units such as arithmetic logic units or multipliers to perform data processing operations





Topic : Datapath





Topic : Datapath



Instruction fetch:-

$AR \leftarrow PC$

$DR \leftarrow M[AR]$

$IR \leftarrow DR, PC \leftarrow PC + 1$

operation

$R1 \leftarrow R2 + R3$

$S \leftarrow R2$

$T \leftarrow R3$

$AC \leftarrow S + T$

$R1 \leftarrow AC$

Total 7 cycles



1

5

1

Total
4 cycles

Assume,

each micro-operation takes 1 CPU cycle but mem.

access micro-operation takes 5 CPU cycle time.



Topic : Datapath



Instruction fetch:-

$AR \leftarrow PC \Rightarrow PC_{out}, AR_{in}$

$DR \leftarrow M[AR] \Rightarrow AR_{out}, Mem_{read}, DR_{in}$

$IR \leftarrow DR, PC \leftarrow PC + 1 \Rightarrow DR_{out}, IR_{in}, PC_{inc}$

operation

$R1 \leftarrow R2 + R3$

$S \leftarrow R2$

$T \leftarrow R3$

$AC \leftarrow S + T$

$R1 \leftarrow AC$



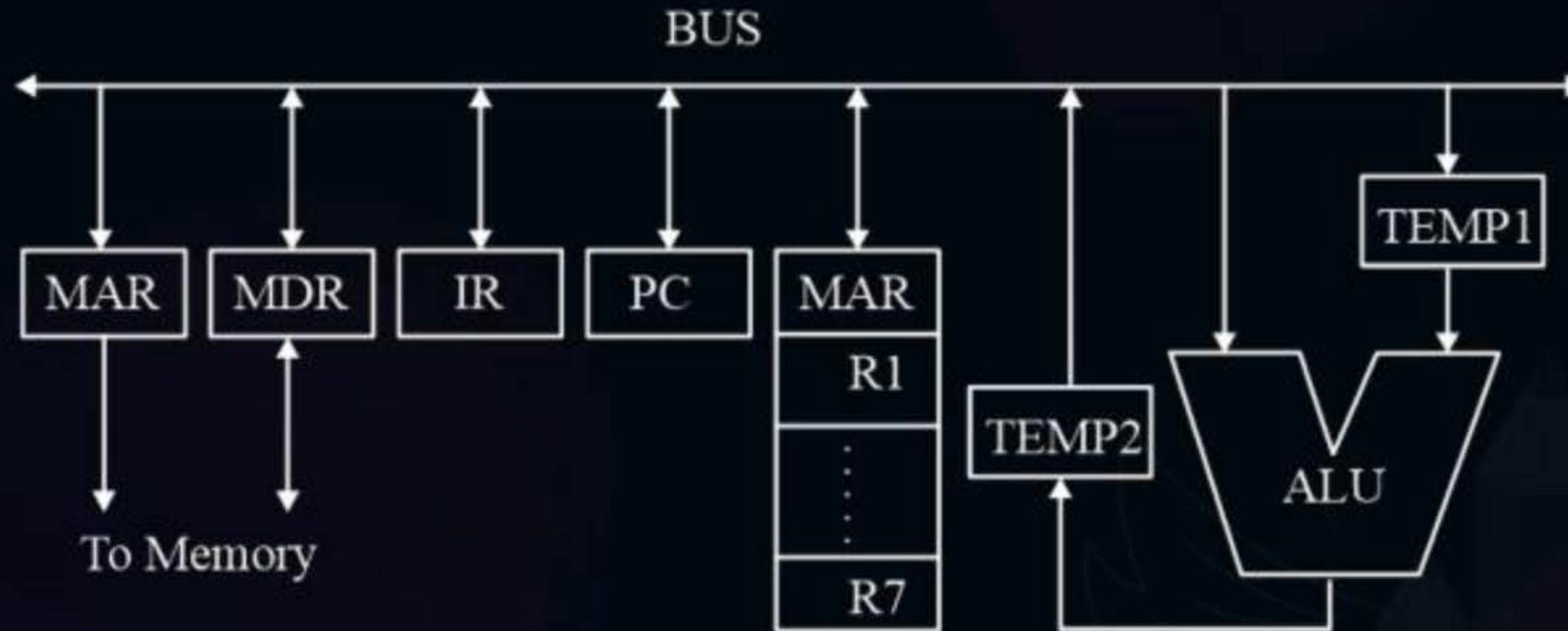
Topic : Control Unit



It generates control signals and sends them to all components.
Components perform operation according to those signals



#Q. Consider the following data path diagram



Consider an instruction: $R0 \leftarrow R1 + R2$. The following steps are used to execute it over the given data path. Assume that PC is incremented appropriately. The subscripts r and w indicate read and write operations, respectively.

1. $R2_r, TEMP1_r, ALU_{add}, TEMP2_w$
2. $R1_r, TEMP1_w$
3. PC_r, MAR_w, MEM_r
4. $TEMP2_r, R0_w$
5. MDR_r, IR_w

Which one of the following is the correct order of execution of the above steps

A 2, 1, 4, 5, 3

B 1, 2, 4, 3, 5

C 3, 5, 2, 1, 4

D 3, 5, 1, 2, 4



2 mins Summary



Topic

CPU

Topic

MIPS

Topic

Data Path



Happy Learning

THANK - YOU

